Cube-Sum Permutations: 66 Integer Identities and Four Algebraic Templates

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Abstract

Let $P(n)$ count permutations p of $\{1,\dots,n\}$ for which every $i+p(i)$ is a perfect cube. We prove $P(n) > 0$ for every $n \geq 173$ from 66 exceptional integer endpoint identities (885 reflected blocks) and four general algebraic block-rearrangement templates giving sixteen further seeds. An elementary suffix-reflection rule and a cubic-gap inequality extend the construction to all later lengths. An explicit predecessor table explains all remaining finite lengths without a matching search, a large permutation archive, or dependent augmenting paths. The finite case distinctions are *not* eliminated: this is a solver-free, hand-checkable certificate rather than a case-free parameterized construction.

1 Two elementary lemmas

Lemma 1 (Boundary cancellation)**.** *Let $0 = b_0 < b_1 < \cdots < b_t = n$ and let $c_1, \dots, c_t$ be positive integer cubes. Set $L_j = c_j - b_j$ and $R_j = c_j - b_{j-1}$. Assume $1 \leq L_j < R_j \leq n+1$. Then $p(i) = c_j - i$ for $b_{j-1} < i \leq b_j$ is a cube-sum permutation if and only if*

$$\boxed{\{1, R_1, \dots, R_t\}_{\text{multi}} = \{n+1, L_1, \dots, L_t\}_{\text{multi}}.}$$

Proof. Each input block maps bijectively onto the integer interval $[L_j, R_j)$. At an integer x, the jump in the total interval-coverage count is the multiplicity of x as a left endpoint minus its multiplicity as a right endpoint. The displayed identity makes that jump 1 at $x = 1$, -1 at $x = n+1$, and zero everywhere else. Thus coverage is exactly one at every $x \in [1, n]$. Reversing this argument proves necessity. □

Lemma 2 (Growing cubic suffix)**.** *Suppose $0 \leq m < n$, $m + n + 1 = c$ is a cube, and a cube-sum permutation on $[1, m]$ exists (the $m = 0$ permutation is empty). Keep the existing permutation on $[1, m]$ and set $p(i) = c - i$ for $m < i \leq n$. Then p is a cube-sum permutation on $[1, n]$.*

Proof. The new reflection is a bijection from $[m+1, n]$ to itself, and each new sum is c. □

2 Four algebraic block families and 66 exceptional identities

Lemma 3 (Reflected-block rearrangement)**.** *Choose positive input-block lengths $\ell_1, \dots, \ell_t$ of sum n and an arbitrary output order π of these t blocks. Let A_j be the sum of input lengths before block j, and let B_j be the sum of output lengths before block j in order π. If all numbers*

$$C_j = 1 + A_j + B_j + \ell_j$$

are perfect cubes, then on input block j the reflection $p(i) = C_j - i$ gives a cube-sum permutation of $[1, n]$.

Proof. The input block is $[A_j + 1, A_j + \ell_j]$, which maps exactly to the prescribed output block $[B_j + 1, B_j + \ell_j]$. Those output blocks partition $[1, n]$ by construction. Each index sum is C_j. □

Family I (four blocks). Let $U < V$ be integer cubes and put

$$a = 2U - V - 1, \qquad b = 2V - 3U + 1.$$

If $a, b > 0$, use input lengths (a, b, a, b) and output order $(3, 4, 1, 2)$. Lemma 3 gives the cubic sums (U, V, U, V) and length $2(V - U)$. The pairs $(U, V) = (64, 125), (125, 216), (216, 343)$ yield lengths $122, 182, 254$ respectively.

Family II (five blocks). For cubes $X < Y < Z$, define

$$a = 2Y - Z - 1, \quad b = X + 2Z - 4Y + 1, \quad c = Y - X.$$

If $a, b, c > 0$, the lengths (a, b, c, a, c) in output order $(4, 2, 5, 1, 3)$ produce sums (Y, X, Z, Y, Z) and total length $2Y - X - 1$. With $(Y, Z) = (64, 125)$ and $X = 27, 8$, the results are $n = 100, 119$.

Family III (six blocks). For integer cubes X, W, Y, Z, put

$$\begin{aligned} a &= 2Y - Z - 1, & b &= X - 2a - 1, \\ c &= W + 2a - 2X + 1, & d &= Y + X - W - 2a - 1. \end{aligned}$$

If these are positive, choose lengths (a, b, c, d, a, d) and output order $(5, 2, 3, 6, 1, 4)$. The cubic sums are (Y, X, W, Z, Y, Z) and the length is $2Z - 2Y + X - W$. Taking $(Y, Z) = (64, 125)$, the choices $(X, W) = (8, 64), (27, 64), (8, 27)$ give $n = 66, 85, 103$, respectively.

Family IV (twelve blocks and three binary switches). Let X, T, Q be integers and put

$$\begin{aligned} u &= X - 7, & v &= T - 2X + 7, & w &= X + 27 - T, \\ z &= Q + 2T - 2X - 121, & h &= X + 152 - T - Q. \end{aligned}$$

Assume $u, v, z \geq 0$ and $w, h > 0$. Use input lengths

$$(3, u, v, w, 3, w, z, h, 3, 27, 3, h)$$

and output block order

$$(11, 2, 3, 6, 9, 4, 7, 12, 5, 10, 1, 8).$$

The reflected-block lemma computes the twelve nominal index-image sums as

$$(125, X, T, 64, 125, 64, Q, 216, 125, 216, 125, 216).$$

When the second, third, or seventh block is nonempty, its nominal sum X, T, or Q, respectively, must be a cube. Any zero-length block is omitted from both orders and imposes no cube condition. The total length is

$$n = 276 + X - T - Q.$$

Take $X \in \{7, 8\}$, choose $T \in \{2X - 7, 27\}$, and independently choose $Q \in \{121 - 2T + 2X, 125\}$. All eight resulting length choices satisfy the positivity and cube conditions. The 2^3 distinct lengths are

$$\boxed{131, 132, 150, 151, 155, 156, 174, 175}.$$

This is one symbolic rearrangement construction with three optional empty blocks, not eight separate finite endpoint certificates.

Lemma 4 (Affine-word rigidity)**.** *Fix a finite list of integer cubes $C_1, \ldots, C_t$. Suppose all breakpoints $b_j(n)$ are affine functions of n with $b_0(n) = 0$ and $b_t(n) = n$. The resulting fixed-cube reflection word can be a valid permutation for at most two distinct values of n.*

Proof. Every legal permutation satisfies the sum identity

$$\sum_{j=1}^{t} C_j \big(b_j(n) - b_{j-1}(n)\big) = \sum_{i=1}^{n} (i + p(i)) = n(n+1).$$

Its left side is affine in n; its right side is a monic quadratic. The equality cannot hold for three distinct values. □

The obstruction concerns *fixed* cube sums and affine endpoints only, not varying cube sums or composition of reflection templates.

In total these *four symbolic identities* give sixteen base lengths

$$66, 85, 100, 103, 119, 122, 131, 132, 150, 151, 155, 156, 174, 175, 182, 254$$

without sixteen separate endpoint-table checks.

The remaining 66 rows. The appendix records 66 exceptional rows of the form $n\ b_1 : k_1\ \cdots\ b_t : k_t$. The corresponding cubical sums are $c_j = k_j^3$ and the first boundary is $b_0 = 0$. For every row, the boundaries increase, $1 \le L_j < R_j \le n+1$, and the multiset equality in Lemma 1 holds exactly. These are 885 simple integer-interval identities. Exactly 10 listed rows have $n < 173$ and exactly 56 have $173 \le n \le 364$. Together with the sixteen algebraic-template lengths, there are 22 lower and 60 upper seeds.

The following two lists give every omitted base length without further matching data. For any tabulated predecessor m under cube c, define $n = c - m - 1$ and apply Lemma 2. The lists are ordered by cube, so every predecessor is either a seed or was obtained in an earlier row.

First: 26 reflected lengths below 173.

```
cube     already constructed predecessors m
  8      0
 27      0,7
 64      0,7,19,26
125      0,7,19,26,37,44,56
216      44,56,63,66,68,80,82,85,87,
         100,103,105
```

Together with the 10 exceptional lower seeds and the twelve algebraic-template lower seeds, these supply 48 distinct lengths below 173.

Second: 49 reflected lengths in $[173, 255] \cup [339, 364]$.

```
cube     previously constructed predecessors m
216      0,7,19,26,37
343      0,87,98,100,103,105,110,112,
         115,116,117,118,119,120,121,122,
         123,124,125,126,128,129,130,131,
         132,133,135,147,149,150,151,152,
         155,156,159
512      147,149,150,151,152,155,156,
         159,171
```

The 56 upper exceptional rows, the four upper algebraic-template lengths and these 49 reflections are exactly the 109 integers in $[173, 255] \cup [339, 364]$. There is no circular dependency: each predecessor m is strictly smaller than its constructed length n.

For example, $211 = 343 - 131 - 1$, $340 = 512 - 171 - 1$, and $342 = 343 - 0 - 1$. The first example replaces a previously necessary-looking 53-segment independent row by one suffix reflection.

Lemma 5 (Complete finite interval)**.** *For every integer $173 \le n \le 364$, a cube-sum permutation of $[1, n]$ exists.*

Proof. The explicit arithmetic rows and the two predecessor tables establish the claim for $[173, 255] \cup [339, 364]$. For each of the 83 remaining lengths $256 \le n \le 338$, take $m = 511 - n \in [173, 255]$. Then $m < n$ and $m + n + 1 = 512 = 8^3$, so Lemma 2 applies. □

3 A purely symbolic infinite tail

Theorem 1. *For every integer $n \ge 173$, there is a permutation p of $[1, n]$ such that $i + p(i)$ is a perfect cube for all indices i.*

Proof. The base interval is Lemma 5. For $n \ge 365$, assume the claim for every smaller integer at least 173, and let k^3 be the least integer cube satisfying $k^3 \ge n + 174$. Thus $m = k^3 - n - 1 \ge 173$.

If $365 \le n \le 555$, then $k = 9$ and $k^3 = 729 \le 2n$. For $n \ge 556$, minimality of k implies $k \ge 10$ and $n \ge (k-1)^3 - 173$. Therefore

$$\begin{aligned} 2n - k^3 &\ge 2(k-1)^3 - 346 - k^3 \\ &= k^3 - 6k^2 + 6k - 348 \\ &\ge 112 > 0. \end{aligned}$$

The final polynomial equals 112 at $k = 10$ and increases thereafter: its first difference is $3k^2 - 9k + 1 > 0$. In either case $k^3 \le 2n$ and hence $m < n$. Apply the induction hypothesis at m and Lemma 2. □

Logical scope. The theorem uses explicit finite arithmetic equalities and short symbolic lemmas. The optional script `verify_66_plus_templates.py` checks the 66 exceptional rows, regenerates all sixteen template cases and checks finite closure without matching or other external data. It is not a premise for the proof. A uniform, case-free formula replacing the remaining 66 exceptional rows is *not* claimed.

A The 66 exceptional integer-multiset witnesses

Every token $b : k$ means $p(i) = k^3 - i$ up to endpoint b. For each row, verify the same endpoint balance of Lemma 1.

82: 2:4 5:2 21:3 42:4 61:5 63:4 82:5

116: 2:4 3:3 4:2 8:3 9:5 10:3 11:5 15:3 16:5 17:3 18:5 22:3 23:5 24:3 61:5 63:4 100:5 101:6 102:5 106:6 107:5 108:6 109:5 113:6 114:5 115:6 116:5

118: 1:3 2:2 3:3 4:2 5:3 6:2 7:5 8:3 9:5 10:3 11:5 12:3 14:5 15:3 16:5 17:3 18:5 19:3 21:5 22:3 23:5 24:3 25:5 26:3 98:5 99:6 100:5 101:6 102:5 103:6 105:5 106:6 107:5 108:6 109:5 110:6 112:5 113:6 114:5 115:6 116:5 117:6 118:5

120: 2:4 3:3 4:2 9:5 10:3 16:5 17:3 23:5 24:3 61:5 63:4 100:5 101:6 107:5 108:6 114:5 115:6 120:5

121: 2:4 3:3 9:5 10:3 16:5 17:3 23:5 24:3 61:5 63:4 100:5 101:6 107:5 108:6 114:5 115:6 121:5

123: 2:4 31:5 32:4 61:5 63:4 92:5 93:6 122:5 123:6

125: 3:5 4:2 29:5 30:4 33:5 34:4 90:5 91:6 94:5 95:6 120:5 121:6 124:5 125:6

126: 2:4 5:2 28:5 35:4 61:5 63:4 89:5 96:6 119:5 126:6

127: 1:5 4:2 5:3 6:2 11:5 12:3 18:5 19:3 25:5 26:3 27:5 30:4 31:5 36:4 88:5 91:6 92:5 97:6 102:5 103:6 109:5 110:6 116:5 117:6 118:5 121:6 122:5 127:6

129: 3:5 4:2 8:3 11:5 15:3 18:5 22:3 25:5 30:4 33:5 38:4 86:5 91:6 94:5 99:6 102:5 106:6 109:5 113:6 116:5 121:6 124:5 129:6

173: 7:2 12:3 14:5 19:3 21:5 42:4 103:6 105:5 110:6 112:5 173:6

176: 7:2 12:3 14:5 17:4 19:3 24:5 42:4 103:6 105:5 110:6 115:5 166:6 169:7 176:6

177: 3:5 4:2 8:3 11:5 15:3 18:5 22:3 25:5 30:4 33:5 38:4 91:6 94:5 99:6 102:5 106:6 109:5 113:6 116:5 121:6 124:5 177:6

179: 3:5 11:3 15:4 23:3 30:4 33:5 40:4 48:6 52:4 91:6 94:5 121:6 124:5 163:6 167:7 175:6 179:7

180: 5:3 7:5 12:3 14:5 19:3 21:5 26:3 28:5 35:4 96:6 98:5 103:6 105:5 110:6 112:5 117:6 119:5 180:6

181: 3:5 4:2 29:5 30:4 33:5 34:4 91:6 94:5 95:6 120:5 121:6 124:5 181:6

184: 7:2 31:4 32:6 56:4 159:6 183:7 184:6

185: 21:5 26:3 30:4 51:6 58:4 91:6 112:5 178:6 185:7

188: 2:5 5:4 7:2 30:4 32:6 61:4 91:6 93:5 154:6 157:7 159:6 188:7

194: 7:2 12:3 14:5 19:3 21:5 103:6 105:5 110:6 112:5 194:6

197: 8:5 18:3 55:6 63:4 91:6 99:5 182:6 190:7 197:6

198: 2:4 5:2 28:5 35:4 53:6 61:5 63:4 71:5 96:6 119:5 144:6 162:7 180:6 198:7

199: 2:4 5:2 16:4 22:6 41:4 47:6 58:4 61:5 63:4 66:5 149:6 168:7 174:6 193:7 199:6

200: 26:3 37:4 51:6 73:5 142:6 164:7 178:6 200:7

201: 1:3 12:4 25:3 37:4 50:6 62:5 63:4 141:6 164:7 177:6 190:7 201:6

202: 13:4 25:6 38:4 50:6 63:4 152:6 165:7 177:6 190:7 202:6

203: 12:4 25:6 26:3 37:4 51:6 62:4 152:6 164:7 178:6 189:7 203:6

204: 11:4 15:6 26:3 36:4 51:6 61:5 142:6 163:7 178:6 188:7 204:6

205: 2:4 5:2 21:3 42:4 46:6 61:5 63:4 78:5 137:6 169:7 173:6 205:7

206: 7:2 18:3 19:4 20:6 44:4 45:6 55:5 56:4 79:5 136:6 146:7 147:6 171:7 172:6 206:7

228: 2:4 3:2 4:3 5:2 22:6 23:3 41:5 58:4 61:5 63:4 101:5 114:6 132:7 149:6 228:7

229: 2:4 5:2 22:6 41:5 58:4 61:5 63:4 102:5 113:6 132:7 149:6 229:7

231: 7:2 20:6 43:5 56:4 104:5 111:6 134:7 147:6 231:7

233: 7:2 9:6 18:4 43:5 45:6 70:5 81:6 106:5 109:6 134:7 136:6 170:7 172:6 197:7 208:6 233:7

234: 16:6 36:5 52:6 72:5 88:6 108:5 127:7 143:6 163:7 179:6 199:7 215:6 234:7

235: 7:2 9:6 17:3 43:5 45:6 79:5 81:6 107:5 134:7 136:6 170:7 172:6 206:7 208:6 235:7

236: 7:2 18:6 43:5 45:6 54:4 79:5 81:6 106:5 134:7 145:6 170:7 172:6 206:7 208:6 236:7

238: 7:2 20:6 43:5 56:4 104:5 134:7 147:6 238:7

240: 2:4 5:2 22:6 41:5 58:4 61:5 63:4 102:5 132:7 149:6 240:7

241: 2:4 3:2 4:3 5:2 22:6 23:3 41:5 58:4 61:5 63:4 101:5 132:7 149:6 241:7

243: 25:4 61:5 63:4 86:6 99:5 190:7 213:6 243:7

245: 27:4 61:5 88:6 97:5 188:7 215:6 245:7

246: 2:6 24:3 28:6 36:5 38:4 60:5 61:6 63:4 64:6 96:5 127:7 129:6 151:7 155:6 187:7 188:6 190:7 191:6 246:7

247: 2:6 3:2 4:3 5:2 22:6 23:3 29:6 36:5 38:4 41:5 58:4 59:5 61:6 63:4 65:6 95:5 127:7 129:6 132:7 149:6 150:7 156:6 186:7 188:6 190:7 192:6 247:7

248: 2:6 5:2 30:6 36:5 38:4 41:5 58:4 61:6 63:4 66:6 94:5 127:7 129:6 132:7 157:6 185:7 188:6 190:7 193:6 248:7

249: 1:3 6:2 31:6 42:5 57:4 62:6 63:4 67:6 93:5 133:7 158:6 184:7 189:6 190:7 194:6 249:7

250: 7:2 32:6 43:5 56:4 68:6 92:5 134:7 159:6 183:7 195:6 250:7

251: 7:2 19:3 33:6 55:5 69:6 91:5 146:7 160:6 182:7 196:6 251:7

252: 7:2 9:6 18:4 34:6 43:5 45:6 54:5 81:6 90:5 134:7 136:6 145:7 161:6 170:7 172:6 181:7 208:6 252:7

253: 2:4 24:3 25:6 27:4 28:6 35:4 60:5 64:6 86:5 88:6 89:5 151:7 152:6 154:7 155:6 187:7 191:6 213:7 215:6 253:7

339: 6:3 20:6 23:4 26:3 172:6 302:8 316:7 322:8 339:7

341: 1:3 21:7 62:6 63:4 189:6 190:7 317:8 341:7

343: 16:5 30:4 46:6 47:4 91:6 107:5 168:6 326:8 342:7 343:8

344: 40:7 42:6 82:5 133:6 167:7 169:8 173:7 175:6 209:7 302:8 342:7 344:8

345: 7:2 166:7 176:8 335:7 345:8

346: 8:5 49:7 63:4 74:6 99:5 165:6 177:8 190:7 201:6 218:7 346:8

347: 9:3 37:4 46:6 51:4 161:6 164:7 330:8 342:7 347:8

348: 10:6 36:5 46:6 52:5 101:6 127:7 137:6 163:7 270:8 342:7 348:8

349: 46:7 53:4 71:6 78:5 162:6 169:8 198:6 205:7 215:6 349:8

350: 9:7 17:4 26:3 54:4 161:6 169:8 178:6 350:8

351: 7:2 19:3 55:7 69:6 97:5 160:6 188:8 196:6 224:7 351:8

353: 6:3 7:2 20:6 25:3 43:5 57:4 116:6 134:7 158:6 285:8 316:7 353:8

354: 5:7 21:4 22:3 58:4 157:6 169:8 173:7 174:6 338:8 342:7 354:8

357: 7:2 19:3 154:7 188:8 323:7 357:8

358: 6:3 7:2 43:7 62:4 85:5 153:6 176:8 189:6 212:7 358:8

363: 4:7 6:6 21:4 42:6 67:5 133:6 148:7 169:8 173:7 300:8 342:7 363:8